



DARK STORES



OUR PROJECT

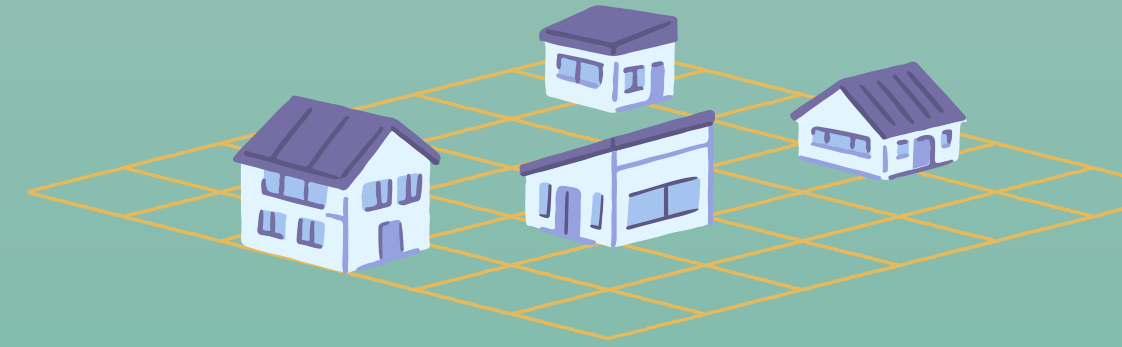


Who We Are?

- An AI-powered decision support system for selecting the best Dark Store locations.
- Combines GIS and artificial intelligence to optimize site selection.
- Uses real-world spatial and demographic data.
- Analyzes population density, road accessibility, nearby services, delivery coverage, and competitor locations.
- Evaluates and ranks candidate locations to recommend the most suitable sites.

PROBLEM STATEMENT

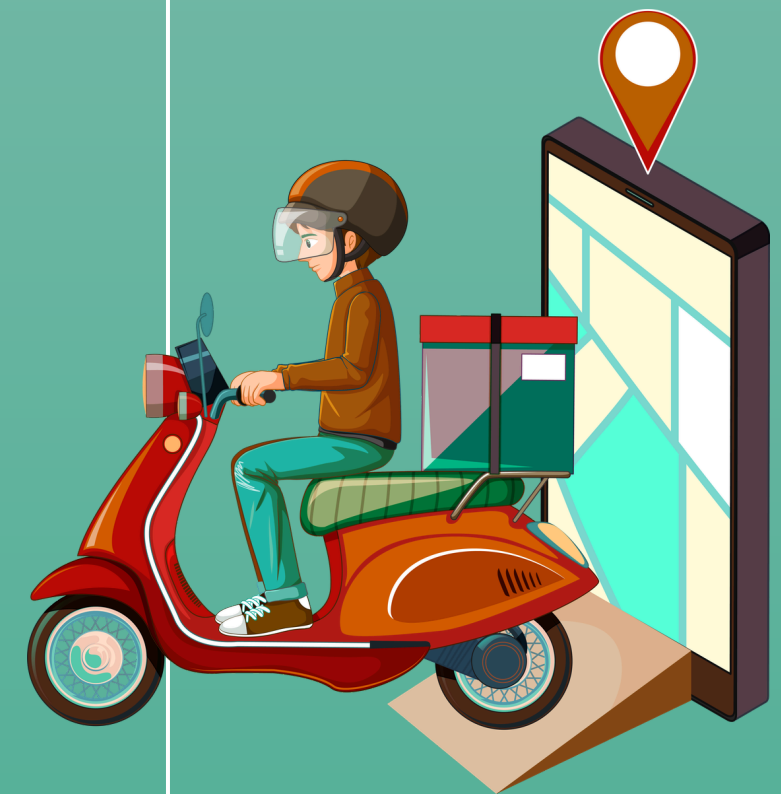
- Choosing the right location for a Dark Store is a complex decision.
- Manual site selection is time-consuming and often inaccurate.
- Companies must analyze multiple geographic and operational factors.
- Poor location selection increases delivery time, operational costs, and reduces customer satisfaction.



PROJECT VALUE



- Transforms complex geographic and operational data into clear, data-driven recommendations.
- Supports strategic business expansion through intelligent location planning.
- Improves last-mile delivery efficiency.
- Reduces operational costs.
- Increases customer coverage with shorter delivery times.
- Enables faster and more accurate decision-making.
- Provides a scalable and objective approach to Dark Store planning.
- Supports business growth and enhances customer satisfaction.



DATASET

Dataset	Description
Population Data	Population distribution and density used to estimate customer demand.
Road Network	Road network data used for accessibility and delivery route analysis.
Points of Interest (POIs)	Nearby services and facilities used to evaluate location suitability.
Competitor Locations	Existing supermarkets and competitors used to avoid saturated areas.
Extracted Features	Processed spatial features generated during data preprocessing and feature engineering.
Final Analysis Results	Ranked candidate locations and final evaluation results generated by the system.



ALGORITHMS

XGboost

Model Overview

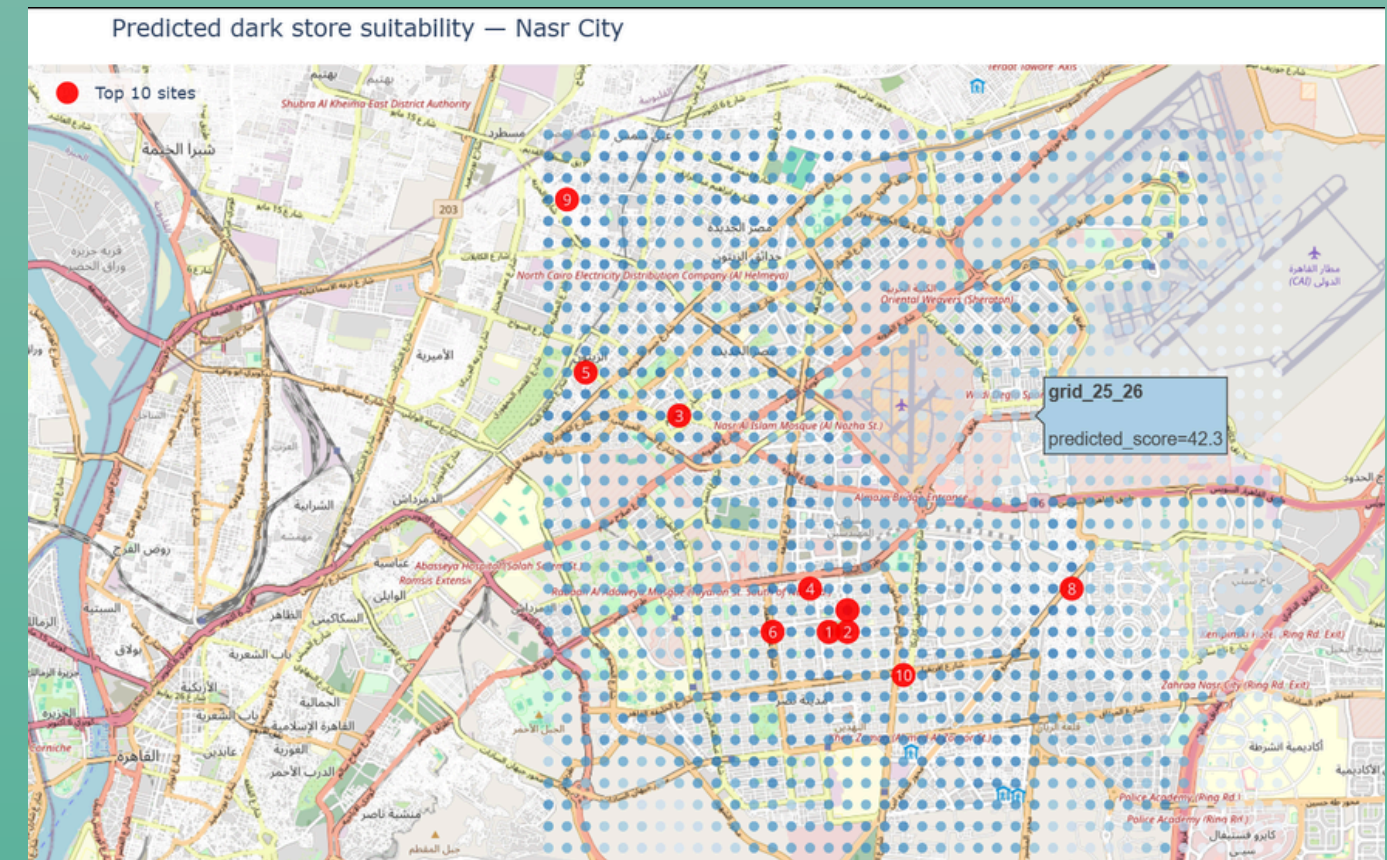
- Built an XGBoost regression model to predict and rank dark store locations across 1,560 grid cells in Nasr City.
- Features included POI density, competitor proximity, road density, and population estimates.
- Evaluated using R^2 , MAE, MSE, and Cross-Validation.

How It Handles Missing Data

- XGBoost handles missing values natively — no manual imputation needed.
- During training, it learns the best default direction for missing values automatically.
- Useful for edge grid cells where POI or road data was incomplete.

How It Handles Overfitting

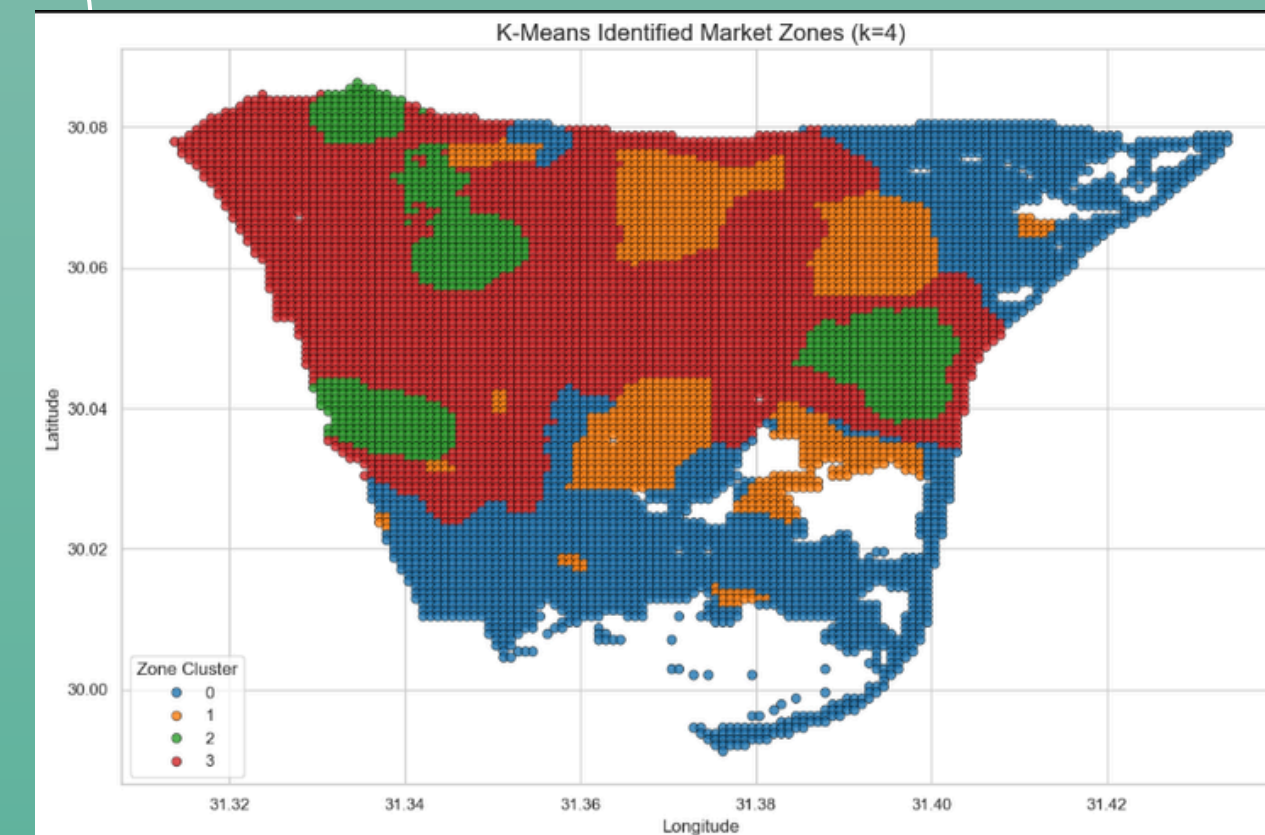
- Regularization (L1 & L2): Penalizes overly complex trees.
- Learning rate: Shrinks each tree's contribution so the model learns gradually.
- Subsampling: Trains on random subsets of rows and columns each iteration.
- Max depth: Limits tree size to avoid memorizing the training data.



ALGORITHMS

K-Means clustering

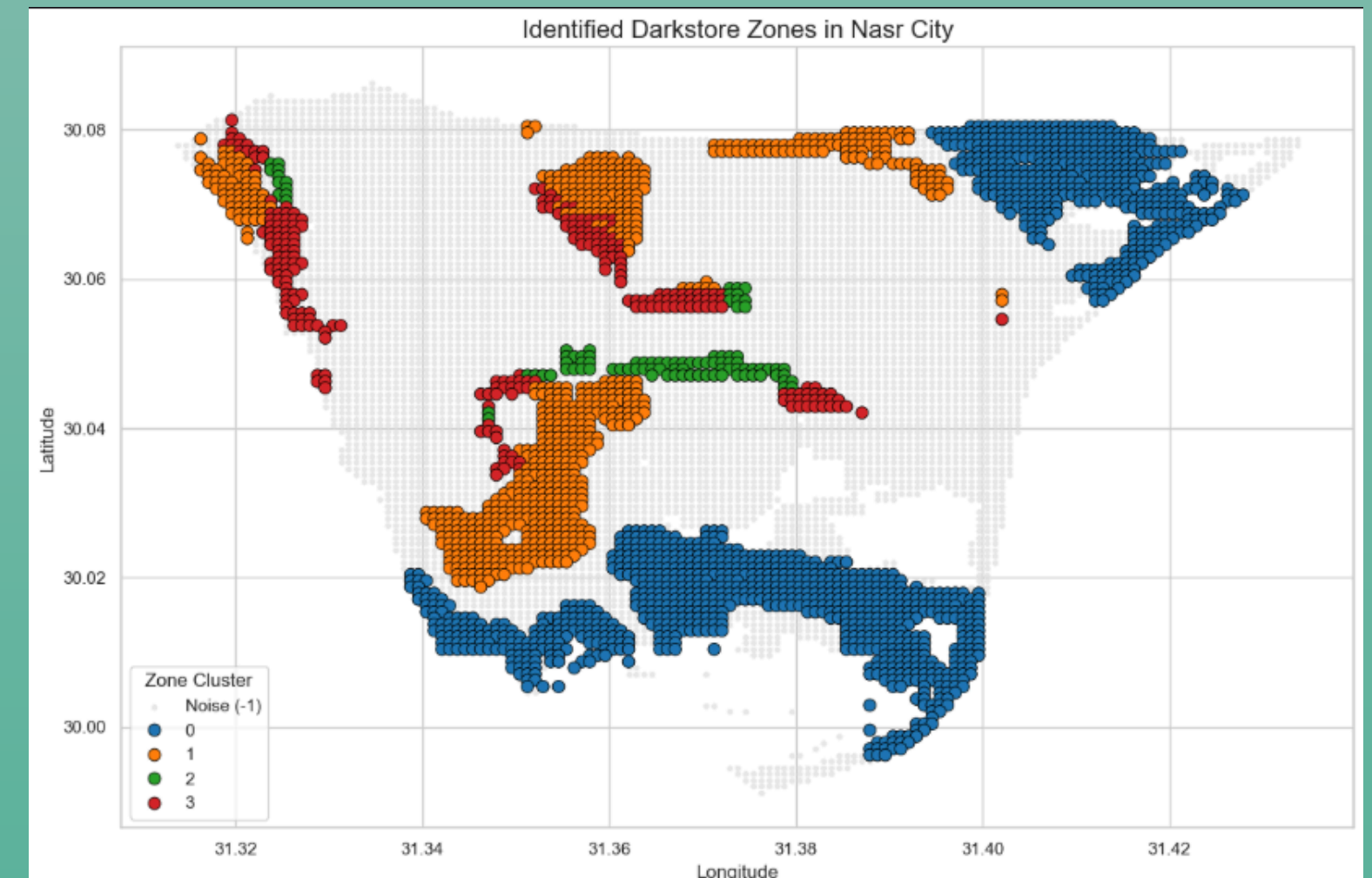
- OPTIMIZATION: EVALUATED CLUSTER COUNTS FROM 2 TO 8 TO FIND THE MOST BALANCED GROUPING. OPTIMIZATION: EVALUATED CLUSTER COUNTS FROM 2 TO 8 TO FIND THE MOST BALANCED GROUPING.
- BEST MODEL: 4 CLUSTERS PROVIDED THE PEAK PERFORMANCE.
- SCORE: ACHIEVED A SILHOUETTE SCORE OF 0.3762.
- OUTPUT: CATEGORIZED THE ENTIRE DATASET INTO FOUR DISTINCT URBAN PROFILES, WITH THE ZONES DISTRIBUTED ACROSS GROUPS OF 2501, 1134, 844, AND 3624.



ALGORITHMS

DBScan clustering

- PARAMETERS: CONFIGURED WITH AN EPSILON OF 0.3 AND A MINIMUM OF 50 SAMPLES TO PRIORITIZE HIGH-DENSITY REGIONS.
- SCORE: ACHIEVED A HIGHER SILHOUETTE SCORE OF 0.4624 STRICTLY ON THE VALID, CLUSTERED ZONES.
- OUTPUT: IDENTIFIED 4 HIGHLY VIABLE DENSE CLUSTERS REPRESENTING 2,270 TOTAL ZONES.
- NOISE HANDLING: SUCCESSFULLY FILTERED OUT 5,833 OUTLIER ZONES THAT LACK THE DENSITY OR ACCESSIBILITY NEEDED FOR A FACILITY..



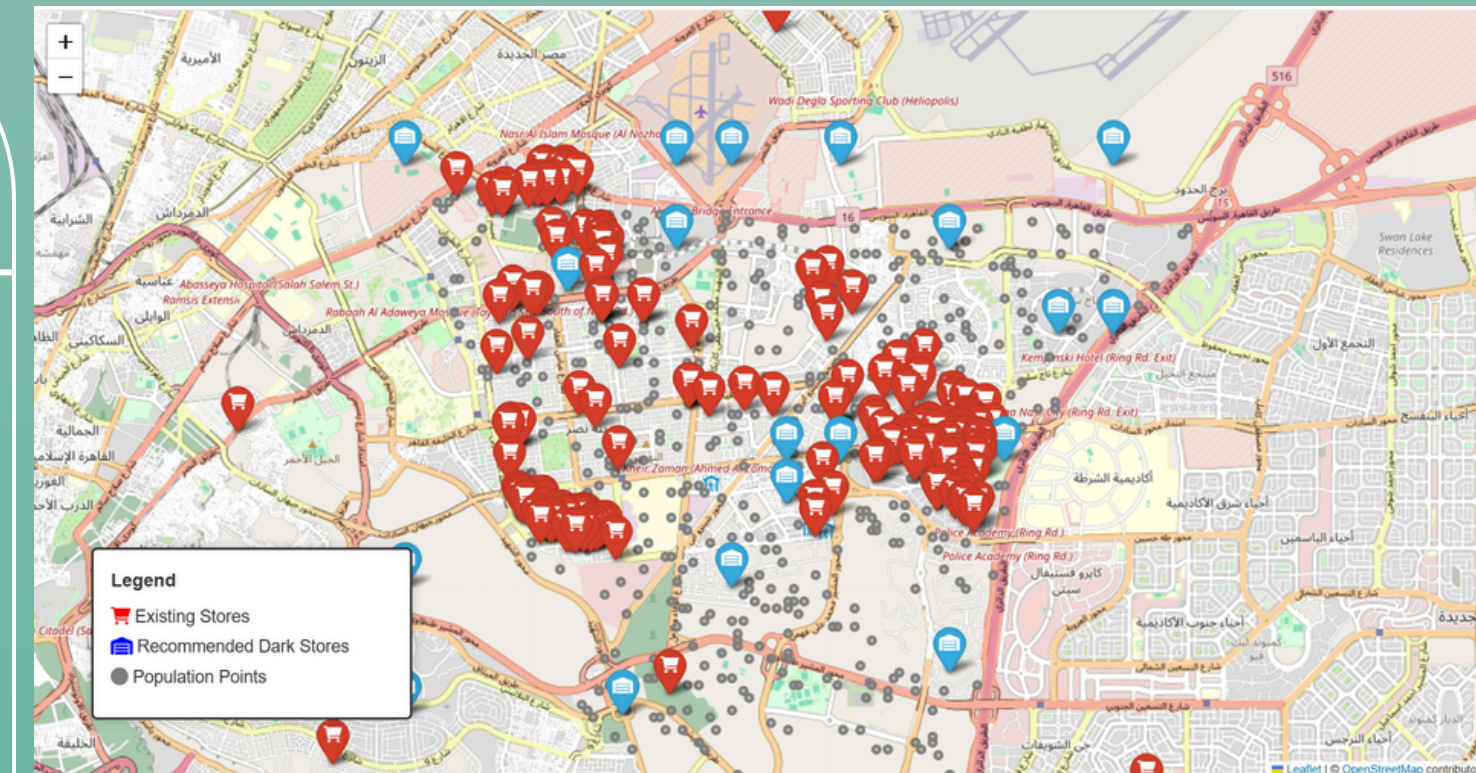
ALGORITHMS

Logistic Regression.

Model Overview

Why Was It Not Used in the Final System?

- The model was designed as a classification model, producing only "Suitable" or "Not Suitable" predictions based on labeled data.
- However, the goal of our project was not simply to classify locations, but to identify and rank the best locations for opening new Dark Stores.
- Therefore, the Machine Learning model was not the most appropriate solution for our project, and we adopted a GIS-based location ranking approach that better achieved our project objectives.



```
### Dark Store #18.0
- **Coordinates**: (30.086250, 31.379031)
- **Population Coverage**: 0.23
- **Accessibility Score**: 0.1662
- **POI Proximity Score**: 0.0365
- **Competition Penalty**: 0.0011
- **Delivery Time Statistics**:
  - Minimum: 6.00 minutes
  - Maximum: 19.97 minutes
  - Mean: 11.09 minutes
  - Median: 10.64 minutes
```

```
### Dark Store #19.0
- **Coordinates**: (30.086250, 31.419832)
- **Population Coverage**: 0.01
- **Accessibility Score**: 0.0370
- **POI Proximity Score**: 0.0103
- **Competition Penalty**: 0.0000
- **Delivery Time Statistics**:
  - Minimum: 6.00 minutes
  - Maximum: 20.00 minutes
  - Mean: 14.22 minutes
  - Median: 14.86 minutes
```

ALGORITHMS

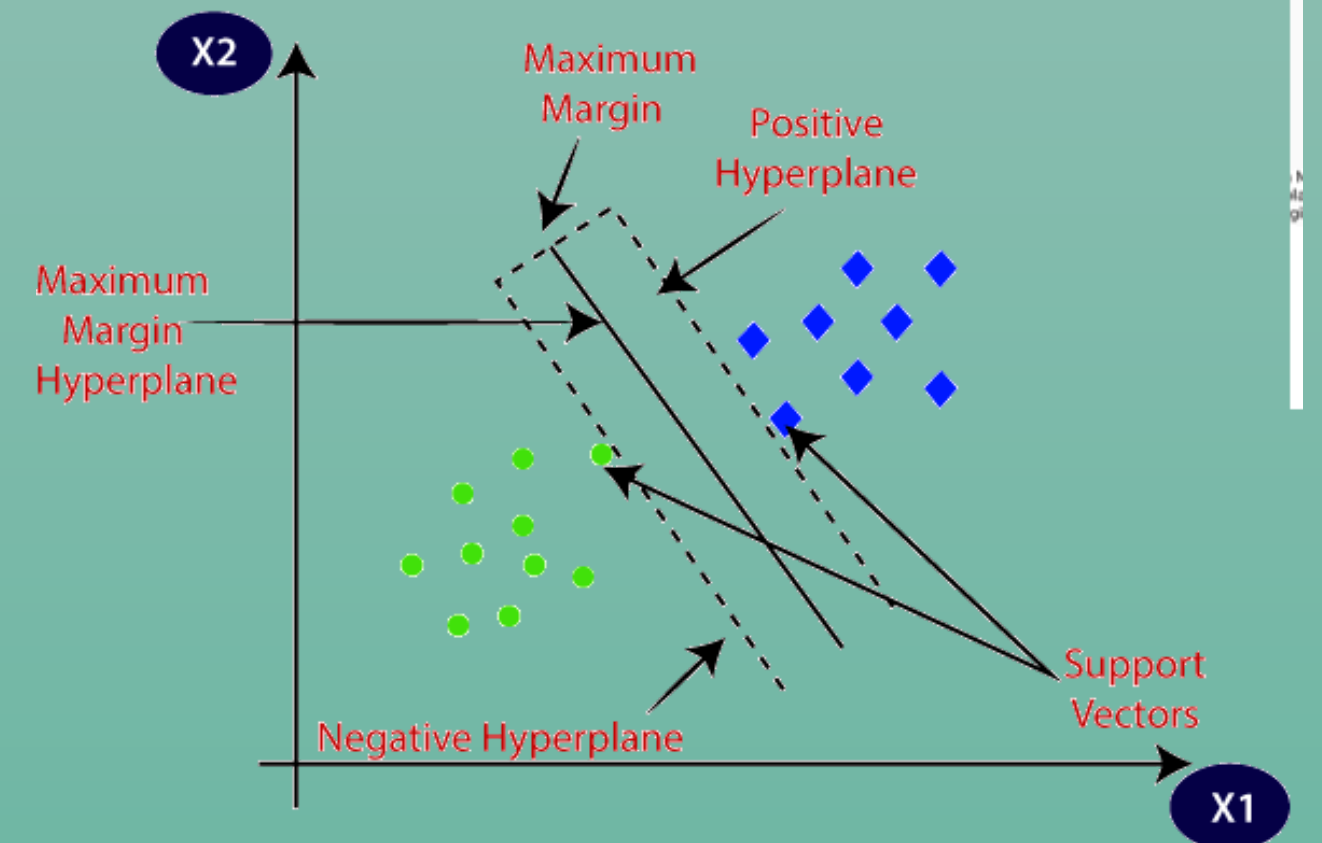
SVM

Model Overview

- Trained and evaluated Support Vector Machines (SVM) across multiple datasets including coordinates, POIs, and population density.
- Used spatial features as inputs, leveraging SVM for its theoretical strength in small-row data, but scaled poorly on our larger data layers.

Why Was It Not Used in the Final System?

- The model was mathematically constrained by dataset sizes, leading to severe performance degradation and underfitting when forced to process our largest datasets.
- However, the goal of our project was not simply to train size-restricted algorithms on isolated layers, but to holistically scale and rank the best locations for opening new Dark Stores.



```
model = Pipeline([
    ("scaler", StandardScaler()),
    ("reg", MultiOutputRegressor(SVR()))
])

param_grid = {
    "reg_estimator_c": [0.1, 1, 10, 100],
    "reg_estimator_kernel": ["linear", "rbf"],
    "reg_estimator_gamma": ["scale", "auto"]
}

# =====
# Training Loop (Grid Search CV)
# =====
model_name = "SVM"
print(f"\nTraining {model_name} via Grid Search for dataset: {file_path.name}...")

grid_search = GridSearchCV(
    estimator=model,
    param_grid=param_grid,
    scoring="neg_root_mean_squared_error",
    cv=cv,
    n_jobs=-1,
    verbose=1
)

start_train = time.time()
grid_search.fit(X_train, y_train)
train_time = time.time() - start_train

best_model = grid_search.best_estimator_
best_models[dataset_key][model_name] = best_model
```

GENETIC ALGORITHM

1- INITIALIZATION

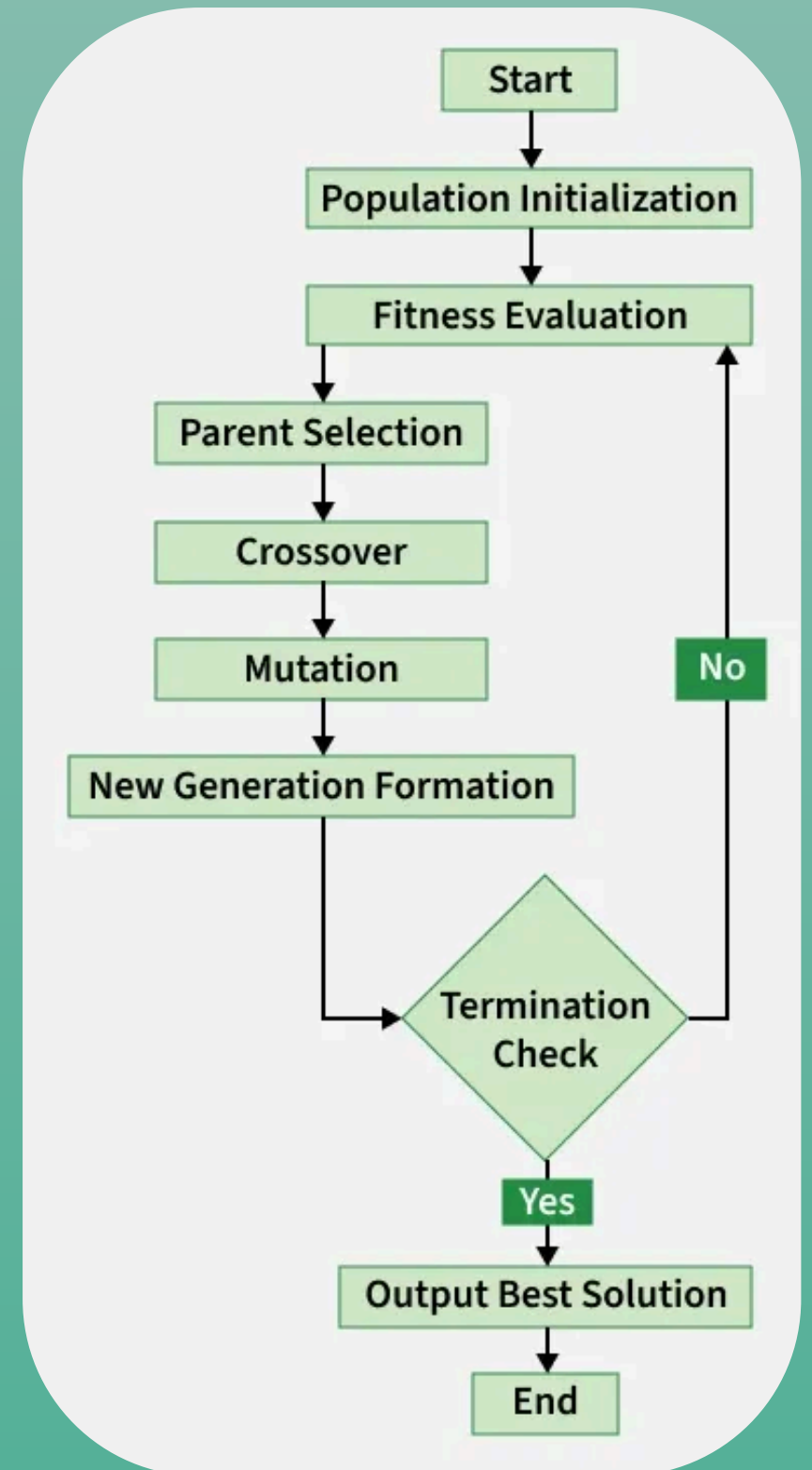
2- FITNESS EVALUATION

3- SELECTION (SURVIVAL OF THE FITTEST)

4- CROSSOVER (MATING)

5- MUTATION (EVOLUTIONARY TWISTS)

6- ITERATION



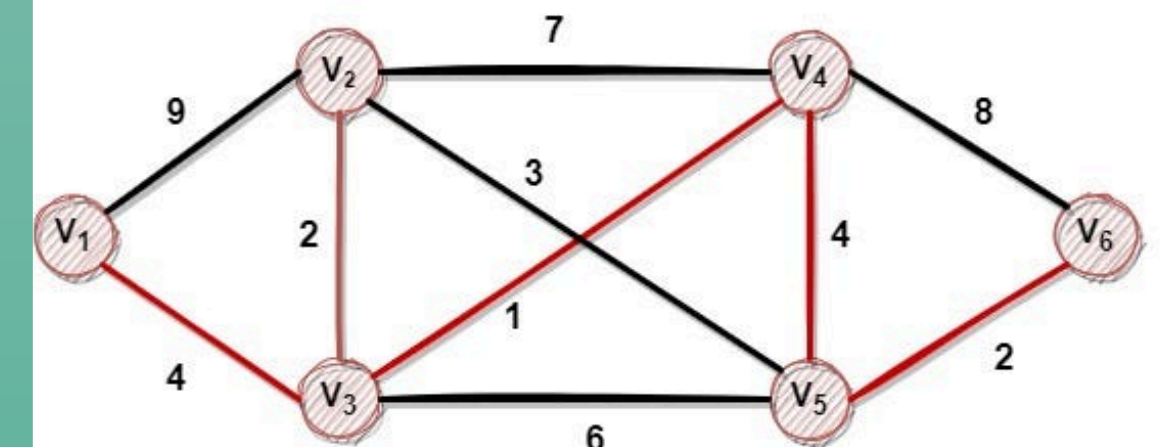
DATA ENGINEERING

1- SPATIAL TRANSLATION (WGS84 → EPSG:32636)

2- SPATIAL TREES (KDTREES)

3- DIJKSTRA ROUTING ALGORITHM

Dijkstra's Algorithm



Path: $V_1 - V_3 - V_4 - V_5 - V_6$

Total Cost: 11

SYSTEM PREVIEW

live demo

project reports

project Diagrams



FUTURE WORK



- Expand the system to cover all governorates across Egypt instead of a single study area.
- Develop a scalable platform that can be deployed globally.
- Integrate real-time traffic, weather, and live demand data for more accurate recommendations.
- Provide dynamic location recommendations based on changing business and customer needs.
- Support multiple companies with customizable business objectives and constraints.
- Deploy the system as a cloud-based AI platform with both web and mobile applications.

CONCLUSION

Our project provides an intelligent, data-driven solution for Dark Store location planning by combining GIS and AI. It helps businesses select optimal locations, improve delivery efficiency, reduce operational costs, and support strategic expansion.





THANK YOU